CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in turning business problems into working agentic AI prototypes built with AI coding agents, and in judging where an agent adds value and where plain automation is enough. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by AI consulting work with business stakeholders in an industrial company.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Sole developer owning an agentic knowledge-extraction system end-to-end, from data ingestion and chunking to prompt engineering and indexing, reporting directly to the CTO and working closely with SAP domain experts.
- Built multi-agent workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) on regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Decided against an LLM for a second, high-volume source stream and built a deterministic, code-based pipeline instead, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Built first versions with Claude Code, custom skills and agentic coding workflows; built an MCP service that gives AI agents millisecond-latency access to the knowledge graph.
- Developed Expert Interface, a web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes themselves.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Claude Code, FastAPI, React, Azure, Docker, Kubernetes, Neo4j, MongoDB**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated their requirements into a concrete solution approach.
- Designed and deployed the resulting agentic AI application with LangGraph, routing each request to vector database search, general LLM inference or summarization.
- Built a React front-end and a Python/FastAPI backend, containerized the stack with Docker and deployed it to AWS with CDK; versioned in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated generation and retrieval evaluation for the PrismaChat RAG assistant with RAGAS and PytrecEval against chemist-provided ground truth, running on AWS (Lambda, SQS, DynamoDB, CDK).
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, retrieval fine-tuning).
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS, Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models on industrial time-series process data to predict synthesis outcomes, and delivered a Streamlit dashboard so chemists could simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load forecasting; the LSTM reached 96% test accuracy on unseen data.

PROJECTS

Project: Order Inbox Agent | Python | Django | TypeScript | Postgres | LLMs

- AI agent for order processing that reads purchase orders from email bodies, PDFs, and Excel files, extracts customer, items, and quantities, matches each line to a product catalog (embeddings plus fuzzy matching), and creates a sales order through a REST API integration with a mock ERP.
- Django backend with Postgres and a TypeScript review screen where a sales rep confirms low-confidence matches before the order is sent; per-customer configuration and a labelled test set of orders run in CI.

TECHNICAL SKILLSET

- **Agentic AI:** AI agents and multi-agent systems (LangGraph, LangChain), LLMs (LiteLLM), MCP, RAG, GraphRAG, Prompt Engineering
- **Rapid Prototyping:** Claude Code, AI coding agents, custom skills, Python, FastAPI, React (beginner), Streamlit
- **Consulting:** stakeholder workflow analysis, requirements to solution approach, advising on where agentic AI fits, Agile
- **Cloud & Delivery:** Azure, AWS (CDK, EC2, Lambda, SQS, DynamoDB), Docker, Kubernetes, Git, GitLab CI/CD
- **Data & ML:** Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search), Scikit-Learn, PyTorch, XGBoost

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)